Zero-dimensional qudit stabilizer codes of odd length from circulant graphs

(Dated: Revised 10 November 2026)

We report four qudit stabilizer codes, $[\![19,0,9]\!]_5$, $[\![25,0,11]\!]_5$, $[\![29,0,12]\!]_5$, and $[\![21,0,10]\!]_7$. All four are circulant graph codes, i.e., stabilizer states of weighted circulant graphs over $\mathbb{F}_q$. When they were found, on 11 October 2026, each exceeded by one the lower bound on the minimum distance then listed in Grassl's tables of quantum codes (8, 10, 11, and 9). The tables have since changed: on 9 November 2026 the four entries list the lower bounds 9, 11, 12, and 10, on pages last modified on 21 and 22 October 2026. The codes stored there are also circulant graph codes, but they are inequivalent to ours, since their numbers of minimum-weight elements differ. We therefore make no claim about the current tables; our claim is the dated comparison and four explicit, exactly verified codes. The minimum distances were certified by an exact Brouwer–Zimmermann-type enumeration over disjoint qudit information sets and, independently, by a rank criterion on all supports of size $d-1$, computed with a separate program. The numbers of minimum-weight stabilizer elements are $A_9 = 12\,920$, $A_{11} = 18\,300$, $A_{12} = 11\,948$, and $A_{10} = 48\,006$. The codes were found by an autonomous multi-agent language-model search that operated against a frozen, test-validated verifier, and we describe the safeguards used to exclude evaluator errors. A possible reason why odd-length zero-dimensional entries lagged behind their even-length neighbors is that $\mathbb{F}_{q^2}$-linear Hermitian self-dual codes exist only at even length, whereas additive graph codes exist at every length. The tabulated upper bounds coincide with the quantum Singleton bound, so all four entries remain open.

I. INTRODUCTION

Quantum error-correcting codes of the stabilizer type [1, 2] extend naturally from qubits to qudits of prime-power dimension [3–5]. A stabilizer code with parameters $[\![n,k,d]\!]_q$ encodes k qudits of dimension q into n and detects all errors of weight less than d. For given n, k, and q, the largest achievable d is unknown in general. Grassl's tables [6] record the best known lower and upper bounds for qubits up to $n = 256$ and for $q \in \{3,4,5,7,8\}$ up to $n = 100$. Each entry has a page with a short construction record, a stabilizer matrix, and a date of last modification; the pages give no credit lines. An explicit code whose minimum distance exceeds a tabulated lower bound improves the table on the day of the comparison, and the claim can be checked exactly. The tables are updated without notice, so such a comparison must be dated.

This paper concerns zero-dimensional codes ($k = 0$). An $[\![n,0,d]\!]_q$ code is a stabilizer state whose nonidentity stabilizer elements all have weight at least d. Such states are equivalent to self-dual additive codes over $\mathbb{F}_{q^2}$ and, up to local Clifford operations, to graph states of weighted graphs [7–11]. They are of interest both as resource states and as building blocks for codes with $k > 0$.

On 10 October 2026 the qudit tables showed a parity effect at $k = 0$. The table listed $[\![18,0,10]\!]_5$ (meeting the quantum Singleton bound) but only $8 \le d \le 10$ for $n = 19$. Similarly, $[\![24,0,11]\!]_5$ sat next to $10 \le d \le 13$ at $n = 25$, and $[\![28,0,12]\!]_5$ next to $11 \le d \le 15$ at $n = 29$. For $q = 7$, $[\![20,0,10]\!]_7$ sat next to $9 \le d \le 11$ at $n = 21$. Since the generic propagation rule $[\![n,k,d]\!]_q \to [\![n+1,k,d]\!]_q$ requires $k > 0$ [2, 4, 5], a good code at even length does not automatically yield one at the next odd length.

Table I lists four circulant graph codes at these odd lengths. On 10 and 11 October 2026 each exceeded the tabulated lower bound by one. This is no longer the case. On 9 November 2026 the tables list the lower bounds 9, 11, 12, and 10 for the four entries, on pages last modified on 21 and 22 October 2026 (Sec. IV E). The codes stored there are circulant graph codes as well, with different first rows, and they are inequivalent to ours. The relation between the table's codes and ours is not known to us. Our claim is therefore limited to the dated comparison and to the four explicit codes with their exact verification; we do not claim to improve the current tables. Section II recalls the graph-code formalism and explains the parity effect. Section III presents the codes. Section IV describes the exact verification and its validation. Section V describes how the codes were found, and Section VI discusses open problems.

TABLE I. The four codes. "Oct." gives the lower and upper bounds in Ref. [6] on 10 October 2026 (re-checked online on 11 October 2026); "Nov." gives them on 9 November 2026, after the table pages were modified on 21 and 22 October 2026 (Sec. IV E). The upper bounds equal the quantum Singleton bound $\lfloor n/2 \rfloor + 1$. d is the exact minimum distance of our code, c is the first half of the circulant's first row (Sec. III), and "deg" is the vertex degree of the graph.

Code	Oct.	Nov.	d	deg	$c_1 c_2 \cdots c_m$
$[\![19,0,9]\!]_5$	8–10	9–10	9	16	`413014221`
$[\![25,0,11]\!]_5$	10–13	11–13	11	20	`323341240420`
$[\![29,0,12]\!]_5$	11–15	12–15	12	28	`31214214324143`
$[\![21,0,10]\!]_7$	9–11	10–11	10	18	`4416064533`

II. PRELIMINARIES

A. Qudit stabilizer codes

Let q be prime. A Pauli operator on n qudits is specified up to phase by $(\boldsymbol{x}|\boldsymbol{z}) \in \mathbb{F}_q^{2n}$, corresponding to $\bigotimes_i X^{x_i} Z^{z_i}$. Two operators commute if and only if the symplectic form $\langle (\boldsymbol{x}|\boldsymbol{z}), (\boldsymbol{x}'|\boldsymbol{z}') \rangle_s = \boldsymbol{x}\cdot\boldsymbol{z}' - \boldsymbol{z}\cdot\boldsymbol{x}'$ vanishes modulo q. A stabilizer code is defined by an isotropic subspace $S \subseteq \mathbb{F}_q^{2n}$ of dimension $n-k$. Its normalizer is the symplectic complement $N = S^{\perp_s}$, and its minimum distance is $d = \min\{\mathrm{wt}(v) : v \in N \setminus S\}$ for $k > 0$ and $d = \min\{\mathrm{wt}(v) : v \in S \setminus \{0\}\}$ for $k = 0$. Here $\mathrm{wt}(\boldsymbol{x}|\boldsymbol{z}) = |\{i : (x_i, z_i) \neq (0,0)\}|$ is the symplectic weight. The quantum Singleton bound $n - k \geq 2(d-1)$ [4, 12] gives $d \leq \lfloor n/2 \rfloor + 1$ for $k = 0$. Equality holds precisely when every reduced state on $\lfloor n/2 \rfloor$ qudits is maximally mixed, i.e., when the stabilizer state is absolutely maximally entangled (AME) [13–15].

B. Graph codes

Let A be a symmetric $n \times n$ matrix over $\mathbb{F}_q$ with zero diagonal, the weighted adjacency matrix of a graph. The rows of $(I \,|\, A)$ span an isotropic subspace of dimension n, because $\langle (e_u|A_u), (e_v|A_v) \rangle_s = A_{vu} - A_{uv} = 0$. The corresponding stabilizer state (graph state) has generators $K_v = X_v \prod_u Z_u^{A_{vu}}$, and its stabilizer elements are $(\boldsymbol{u} \,|\, \boldsymbol{u}A)$ for $\boldsymbol{u} \in \mathbb{F}_q^n$. Hence

$$d = \min_{\boldsymbol{u} \neq 0} \big|\mathrm{supp}(\boldsymbol{u}) \cup \mathrm{supp}(\boldsymbol{u}A)\big|. \tag{1}$$

Every stabilizer state is local-Clifford equivalent to a graph state [8, 10, 16], so graph codes realize every attainable $[\![n, 0, d]\!]_q$.

A graph is *circulant* if $A_{ij} = a_{(j-i) \bmod n}$. Symmetry requires $a_j = a_{n-j}$, so for odd $n = 2m+1$ the first row is $(0, c_1, \ldots, c_m, c_m, \ldots, c_1)$ and there are q^m circulant graph codes of length n. Identifying $(\boldsymbol{x}|\boldsymbol{z})$ with $\boldsymbol{x} + \omega\boldsymbol{z}$, where $\{1, \omega\}$ is a basis of $\mathbb{F}_{q^2}$ over $\mathbb{F}_q$, a circulant graph code is the additive cyclic code over $\mathbb{F}_{q^2}$ generated by $1 + \omega a(x)$ with $a(x) = \sum_j a_j x^j$. Besides the cyclic shift, a symmetric circulant is invariant under the reflection $j \mapsto -j$, so its symmetry group contains the dihedral group of order $2n$. Danielsen [11] searched circulant graph codes exhaustively for the alphabets $\mathbb{F}_4$, $\mathbb{F}_9$, $\mathbb{F}_{16}$, and $\mathbb{F}_{25}$. For $\mathbb{F}_{25}$ ($q = 5$) the search extends to $n = 18$, where it attains $d = 10$; the length-19 row of that table has no entry for $\mathbb{F}_{25}$. Circulant constructions are also a standard source of good qubit codes [17, 18].

C. Odd versus even length

A $\mathbb{F}_{q^2}$-linear code that is Hermitian self-dual has dimension $n/2$ and therefore requires even n. Such codes give $[\![n, 0, d]\!]_q$ stabilizer codes, and families of $\mathbb{F}_{q^2}$-linear codes (constacyclic, quasi-cyclic, quasi-twisted) are widely used in systematic searches [19, 20]. At odd length the corresponding code must be genuinely additive ($\mathbb{F}_q$-linear but not $\mathbb{F}_{q^2}$-linear). Graph codes are additive and exist at every length.

The table pages for the cells considered here are consistent with this picture, as far as they go. In the copies we saved on 10 and 11 October 2026, the even-length neighbor $[\![18, 0, 10]\!]_5$ is recorded with construction type "Ball" as a quasi-cyclic code with generators $(1, g(x))$ over $\mathbb{F}_{25}$ (page last modified 23 May 2024). The page gives no reference; we take the label to refer to the quantum MDS codes from Hermitian self-orthogonal generalized Reed–Solomon codes of Ball and of Ball and Vilar [21, 22]. The odd-length entries $[\![19, 0, 8]\!]_5$, $[\![25, 0, 10]\!]_5$, and $[\![21, 0, 9]\!]_7$ were recorded as stored generator matrices, on pages last modified in October 2014, and $[\![29, 0, 11]\!]_5$ was derived from a stored $[\![30, 0, 12]\!]_5$ code by shortening and taking a subcode (also October 2014). The pages do not say how the stored matrices were found. The even/odd asymmetry is thus a plausible explanation for the parity effect noted in the Introduction, not an established one. It suggested where an additive family could improve the tables.

III. RESULTS

Each code in Table I is the circulant graph code over $\mathbb{F}_q$ whose adjacency matrix has first row

$$(0,\, c_1, \ldots, c_m,\, c_m, \ldots, c_1), \qquad n = 2m+1, \tag{2}$$

with row i the cyclic right shift of row 0 by i places. The stabilizer is generated by $K_v = X_v \prod_u Z_u^{A_{vu}}$, $v = 0, \ldots, n-1$. Full first rows and explicit stabilizer elements of minimum weight are given in Table II. The minimum-weight elements show that the distances are attained, i.e., the certified lower bounds are tight for these particular codes.

The four codes are invariant under the cyclic shift $v \mapsto v+1$. Since the generator K_v has weight $1 + \deg v$, a graph code with distance d needs every vertex degree to be at least $d-1$. Danielsen showed more: $d-1$ equals the minimum vertex degree over the orbit of the graph under generalized local complementation [11]. Our graphs are dense, with degrees 16–28. The numbers of stabilizer elements of minimum weight, computed in Sec. IV D, are

$$\begin{aligned} A_9 &= 12\,920, \quad A_{11} = 18\,300, \\ A_{12} &= 11\,948, \quad A_{10} = 48\,006 \end{aligned} \tag{3}$$

for $n = 19$, 25, 29, and 21, respectively, and there are no elements of lower weight. For prime n the support of an element of weight less than n is not invariant under any nontrivial cyclic shift, so such an element lies in an orbit of size $n(q-1)$ under shifts and nonzero scalars. Accordingly $A_9 = 76 \times 170$ and $A_{12} = 116 \times 103$. (Elements of

TABLE II. Full first rows $a = (a_0, \ldots, a_{n-1})$ of the circulant adjacency matrices, and one stabilizer element $(\boldsymbol{x}|\boldsymbol{z})$ of minimum weight, with $\boldsymbol{z} = \boldsymbol{x}A$, for each code, written as strings of $\mathbb{F}_q$ digits.

Code	
$[\![19, 0, 9]\!]_5$	$a =$ 0413014221122410314
	$(\boldsymbol{x}\|\boldsymbol{z}) =$ 0200010000001310010 | 0000010000000404122
$[\![25, 0, 11]\!]_5$	$a =$ 03233412404200240421433 23
	$(\boldsymbol{x}\|\boldsymbol{z}) =$ 3000240110242000000040003 | 4000232100320000000030001
$[\![29, 0, 12]\!]_5$	$a =$ 03121421432414334142341241213
	$(\boldsymbol{x}\|\boldsymbol{z}) =$ 0402301010003100000000420004 | 0304102000014400000000114000
$[\![21, 0, 10]\!]_7$	$a =$ 044160645333354606144
	$(\boldsymbol{x}\|\boldsymbol{z}) =$ 600060041600500000020 | 000060012002500500310

weight n can have smaller orbits: for $n = 19$ the element with $\boldsymbol{u} = (1, \ldots, 1)$ is shift invariant.)

Propagation. A pure $[\![n, 0, d]\!]_q$ code yields $[\![n-1, 1, d-1]\!]_q$ by puncturing [2, 4, 5]. For the four codes this gives $[\![18, 1, 8]\!]_5$, $[\![24, 1, 10]\!]_5$, $[\![28, 1, 11]\!]_5$, and $[\![20, 1, 9]\!]_7$, none of which exceeds the table entries of 10 October 2026 (9, 10, 11, and 9, respectively). No further entry is affected.

Status of the entries. The upper bounds in Table I equal the quantum Singleton bound, on both dates. Every entry thus remains open. The entry $[\![21, 0, d]\!]_7$ has $d \in \{10, 11\}$, and a code with $d = 11$ would be an AME(21, 7) stabilizer state. Analogous statements hold for $q = 5$ at $n = 19$, 25, and 29.

Search heuristics. A fixed nonzero vector lies in a uniformly random maximal isotropic subspace of $\mathbb{F}_q^{2n}$ with probability $1/(q^n + 1)$. Hence $E = \sum_{w=1}^{d-1} \binom{n}{w}(q^2 - 1)^w/(q^n + 1)$ is the expected number of nonzero stabilizer elements of weight below d in a uniformly random stabilizer state. As a heuristic we transfer this number to random symmetric circulants. For prime n the elements of weight below n occur in orbits of size $n(q-1)$ under shifts and scalars. Treating the number of such orbits as Poisson with mean $\mu = E/[n(q-1)]$ suggests a success probability of roughly $e^{-\mu}$ per random circulant.

Table III compares this estimate with measured rates. For each cell we drew $c \in \mathbb{F}_q^m$ uniformly at random and decided $d \geq d_{\text{target}}$ exactly, by the rank criterion of Sec. IV D. The estimate is too high at all four cells: by a factor of 1.2 at $n = 19$, 3.5 at $n = 21$ $(q = 7)$, 4.9 at $n = 25$, and 6.0 at $n = 29$. The estimate is therefore not a calibrated prediction. It ignores that a circulant code is far from a uniformly random stabilizer state. One identifiable source of error is the reflection symmetry: if $\boldsymbol{u}$ is symmetric or antisymmetric under $j \mapsto s - j$ for some s, then so is $\boldsymbol{u}A$, so the entries of the $\boldsymbol{z}$ part are not independent and low weights are more likely than in the uniform model. We have not quantified how much of the discrepancy this explains. The agent that found the codes used μ only to rank candidate entries (Sec. V). The four entries it reached have $\mu \leq 7.1$; its sampling runs at entries with $\mu \approx 9$ and $\mu \approx 20$ found nothing.

IV. VERIFICATION

Because the result is a claim about a minimum over $q^n - 1$ stabilizer elements, we treated verification as the primary deliverable. All checks below are exact, and floating-point arithmetic is never used.

A. Validity and dimension

The verifier reads the generators, checks that all entries lie in $\mathbb{F}_q$, and confirms isotropy pairwise. It computes $k = n - \operatorname{rank} S$ by Gaussian elimination over $\mathbb{F}_q$ and the normalizer as the kernel of $(-Z \,|\, X)$, checking $\dim N = n + k$.

B. Exact minimum distance

Classical Brouwer–Zimmermann enumeration [23] uses several disjoint information sets, i.e., coordinate sets on which a generator matrix can be brought to systematic form. For symplectic weight the relevant coordinates are qudits rather than individual columns, as in the algorithm of White and Grassl for additive codes [24]. We use the following variant. For information set j we reduce a generator matrix of N (here $N = S$) to systematic form with all pivot columns inside a set I_j of qudits, which is disjoint from the previous sets. Rows are grouped by the qudit carrying their pivot; a qudit carries one or two pivots. The f_j rows without a pivot in I_j form singleton groups. Every pivot column vanishes in all other rows. Hence, if the coefficient vector of a codeword v is nonzero on a groups, then v is nonzero on at least $a - f_j$ qudits of I_j.

Suppose all codewords whose coefficient vectors activate at most t_j groups have been enumerated for each j. Because the I_j are disjoint, every codeword not yet seen satisfies

$$\operatorname{wt}(v) \;\geq\; \sum_j \max\bigl(0,\; t_j + 1 - f_j\bigr). \tag{4}$$

The enumeration raises the t_j in rounds and stops when the bound (4) reaches the weight of the lightest codeword

TABLE III. First-moment heuristic for random symmetric circulant graph codes at the target distance, and measured success rates. q^m is the number of such codes. "Hits" gives the number of uniformly random circulants with $d \geq d_{\text{target}}$ (exact test) among the number sampled; the interval is the exact 95% binomial confidence interval.

Code	q^m	E	$\mu = E/[n(q-1)]$	$e^{-\mu}$	Hits/samples	Measured rate	95% interval
$[\![19, 0, 9]\!]_5$	2.0×10^6	449	5.9	2.7×10^{-3}	452/200 000	2.3×10^{-3}	$(2.06–2.48) \times 10^{-3}$
$[\![25, 0, 11]\!]_5$	2.4×10^8	714	7.1	7.9×10^{-4}	49/300 000	1.6×10^{-4}	$(1.21–2.16) \times 10^{-4}$
$[\![29, 0, 12]\!]_5$	6.1×10^9	290	2.5	8.2×10^{-2}	41/3 000	1.4×10^{-2}	$(0.98–1.85) \times 10^{-2}$
$[\![21, 0, 10]\!]_7$	2.8×10^8	722	5.7	3.2×10^{-3}	93/100 000	9.3×10^{-4}	$(0.75–1.14) \times 10^{-3}$

TABLE IV. Exact certification with the frozen verifier. "Words" is the number of codewords enumerated up to scalars (the same in both runs). Run 1 is the board submission of 11 October 2026 (seed 1; the thread count was not recorded, the default is 8). Run 2 is a repetition on 10 November 2026 with seed 977 and 8 threads, with the reports kept in the repository. Times are engine wall-clock times on one laptop CPU (Apple M4, 10 cores); the engine was compiled with `cc -O3 -march=native` (Apple clang).

Code	Words	Time (s), run 1 / run 2	Result
$[\![19, 0, 9]\!]_5$	2.4×10^7	0.05 / 0.05	$d = 9$
$[\![25, 0, 11]\!]_5$	3.6×10^9	11.5 / 9.8	$d = 11$
$[\![29, 0, 12]\!]_5$	1.7×10^{11}	559 / 643	$d = 12$
$[\![21, 0, 10]\!]_7$	1.2×10^{10}	28.5 / 44.5	$d = 10$

found outside S (outside $\{0\}$ for $k = 0$). That weight is then exact. Two refinements reduce the work. The first active group's local value is normalized to fix scalar multiples. Qudits on which S contains a weight-one element are removed beforehand, which leaves d unchanged for $k > 0$. Each witness codeword reported by the engine is re-checked independently: it must commute with all generators, lie outside S, and have the stated weight.

Table IV lists the certification runs. For $k = 0$ codes of length n the algorithm uses two disjoint information sets of about $n/2$ qudits each, so the lower bound grows by about two per round. Each certificate was repeated with a different seed, which changes the random order of the qudits and hence the information sets, and the results agreed. The two runs share all code, so the second is a repetition and not independent evidence. Independent evidence is given in Sec. IV D.

C. Validation of the verifier

Before any search began, the verifier was frozen and tested. It reproduces the distances of standard codes: the $[\![5, 1, 3]\!]_2$, Steane, and Shor codes (the last correctly reported as impure), a Golay-derived $[\![23, 1, 7]\!]_2$ code, and five-qudit codes over $\mathbb{F}_3$ and $\mathbb{F}_5$. It rejects invalid inputs, such as noncommuting generators, wrong k, entries outside $\mathbb{F}_q$, and nonprime q. On 500 random stabilizer codes over $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$ with $n \leq 10$, some padded with fixed qudits, it agrees with a brute-force checker that shares no code with it (test suite with the option `--random 500`, repeated and logged on 10 November 2026; the default is 200, and the freeze script uses 300). That checker enumerates Pauli operators by increasing weight and tests commutation and stabilizer membership directly. On 14 best-known codes stored with Ref. [6] ($q \in \{2, 3, 5, 7\}$, up to $[\![60, 2, 16]\!]_2$), it recovers exact distances equal to the tabulated lower bounds. The sources and the compiled binary were then pinned by cryptographic hash. The submission pipeline refused to accept improvements from a modified verifier. The pin of the binary is meaningful only on the machine that built it, because the engine is compiled for the native architecture; on another machine only the source hashes can be compared. A verdict exceeding a tabulated (proven) upper bound would have been flagged as a verifier fault rather than a discovery.

D. Independent checks

Rank criterion, all four codes. Let G be an $n \times 2n$ generator matrix of the stabilizer. For a set W of w qudits, the stabilizer elements supported inside W form a space of dimension $k(W) = n - \operatorname{rank} G_{\overline{W}}$, where $G_{\overline{W}}$ consists of the $2(n-w)$ columns of the qudits outside W. For a graph code this is the dimension of the left kernel of $A[W, \overline{W}]$. An element of weight at most w is supported inside some w-subset. Hence $d \geq w + 1$ if and only if $k(W) = 0$ for all $\binom{n}{w}$ subsets of size w. If this holds for $w = d - 1$, every nonzero element supported inside a d-subset has weight exactly d, and $A_d = \sum_{|W|=d} (q^{k(W)} - 1)$.

A separate program computes $k(W)$ by Gaussian elimination for every subset, without using the cyclic symmetry. It shares no code with the verifier and uses neither information sets nor codeword enumeration. For $w = d - 1$ it visited 75 582, 3 268 760, 34 597 290, and 293 930 subsets for $n = 19$, 25, 29, and 21, respectively, and found $k(W) = 0$ for all of them. For $w = d$ it gave $A_9 = 12\,920$, $A_{11} = 18\,300$, $A_{12} = 11\,948$, and $A_{10} = 48\,006$. This proves $d = 9$, 11, 12, and 10 exactly. The largest case, $n = 29$, took less than four minutes on 8 threads. The input of this check is rebuilt from the c strings of Table I and compared with the candidate files submitted to the verifier.

Direct enumeration, $n = 19$. Every stabilizer element $(\boldsymbol{u}|\boldsymbol{u}A)$ has weight at least $\mathrm{wt}(\boldsymbol{u})$. A third pro-

gram enumerated all 1 475 748 943 coefficient vectors $\boldsymbol{u}$ with $1 \le \mathrm{wt}(\boldsymbol{u}) \le 8$ (first nonzero entry fixed to 1). It found minimum weight 9. Extending the enumeration to $\mathrm{wt}(\boldsymbol{u}) \le 9$ gave the same count $A_9 = 12\,920$.

E. Relation to the tables and to prior work

Table V gives the state of the four table entries on two dates. We keep copies of the pages. On 11 October 2026 the online tables [6] listed the lower bounds 8, 10, 11, and 9, on pages last modified in October 2014. On 9 November 2026 they list 9, 11, 12, and 10, on pages last modified on 21 and 22 October 2026, with construction type "Gra" and the description "cyclic code of length n with generating polynomial $s(x) + w$". In all four cases s has coefficients in $\mathbb{F}_q$ and is palindromic with $s_0 = 0$, and the stabilizer matrix printed on the page spans $\{(\boldsymbol{v}S \,|\, \boldsymbol{v})\}$, where S is the symmetric circulant with first row s. Up to a Fourier transform on every qudit, the table's codes are therefore circulant graph codes, from the same family as ours.

They are not the same codes. No index multiplier $j \mapsto tj$ combined with a nonzero scalar maps s to our first row. More decisively, the rank criterion of Sec. IV D applied to the stabilizer matrices on the pages confirms their distances and gives numbers of minimum-weight elements that differ from ours in all four cases (Table V). Since qudit permutations and local Clifford operations preserve the weight distribution, each of our codes is inequivalent to the table's code with the same parameters.

The pages give no credit line, and the relation between the table's codes and ours is not known to us. We can therefore document only the two table states and their dates. The six neighboring entries that we checked on 9 November 2026 are unchanged: $[\![20, 0, d]\!]_5$ with $9 \le d \le 11$, $[\![21, 0, d]\!]_5$ with $9 \le d \le 11$, $[\![23, 0, d]\!]_5$ with $10 \le d \le 12$, $[\![27, 0, d]\!]_5$ with $11 \le d \le 14$, $[\![31, 0, d]\!]_5$ with $12 \le d \le 16$, and $[\![23, 0, d]\!]_7$ with $10 \le d \le 12$.

Danielsen's exhaustive circulant search [11] stops at $n = 18$ for $\mathbb{F}_{25}$ and does not treat $\mathbb{F}_{49}$. Our literature search found no other construction with these parameters, but it was short, and we cannot rule out codes in sources we did not examine.

V. HOW THE CODES WERE FOUND

The codes were found by an autonomous multi-agent system built on a large language model (Claude Sonnet 5.5, Anthropic). It was set up to test whether open, verifiable problems of this type can be attacked by AI agents without compromising rigor. Five agents ran in parallel as non-interactive coding sessions on one laptop. They coordinated through a shared message board, where they could claim targets, post progress and dead ends, and submit candidates. The agents were assigned different families: qubit quasi-cyclic and cyclic codes, qubit graph codes and seeded local search, qutrit codes, and $q = 5, 7$ codes. The pool of targets comprised 4 295 table entries whose improvement by one could be certified in about ten minutes.

The design principle was *verifier first*. Agents could not modify the verifier, the bound tables, or the board software. A submission was copied read-only and verified, and the verdict was posted by the verifier, not by the submitting agent. Only verifier-posted improvements counted.

The agent assigned to $q = 5, 7$ noted the parity effect of Sec. II C. It formulated the first-moment estimate of Table III, selected odd-length $k = 0$ entries with small μ, and wrote a sampler for random symmetric circulants. The sampler discarded a circulant if some element $(\boldsymbol{u}|\boldsymbol{u}A)$ with $\mathrm{wt}(\boldsymbol{u}) \le 3$ had weight below the target. The others went to the engine's randomized upper-bound screen (random information sets, 20 iterations, then 300 iterations at a higher enumeration level for survivors). The first certified improvement over the table of that day, $[\![19, 0, 9]\!]_5$, was posted six minutes after launch, and the remaining three followed within 13 minutes. Each came from a single sampling run of a few minutes. The other agents' searches, which included exhaustive qubit circulant graph codes for $26 \le n \le 45$, all Hermitian dual-containing cyclic $\mathbb{F}_4$-linear codes of odd length $41 \le n \le 99$, and exhaustive $\mathbb{F}_3$ circulant, bordered-circulant, and negacirculant graph codes for $14 \le n \le 20$ (circulant up to 23), matched but did not beat the tabulated bounds within certifiable range. All four codes were then re-certified and independently checked as described in Sec. IV.

VI. OUTLOOK

On 9 November 2026 the neighboring odd-length $k = 0$ entries listed in Sec. IV E are unchanged, and additive constructions may improve some of them. The entry $[\![31, 0, d]\!]_5$ (lower bound 12) has $\mu \approx 9$ for $d = 13$ in Table III's heuristic. In view of the measured rates the true success rate of random circulants is probably well below e^{-9}; a seven-minute sampling run during the search (2163 screened circulants) found no candidate. Exact certification at $n = 31$ is estimated at about three quarters of an hour with the present algorithm; the rank criterion of Sec. IV D with the cyclic symmetry would be cheaper. For larger lengths certification rather than search becomes the bottleneck. Exploiting the cyclic automorphism group in the enumeration could reduce the cost by roughly a factor of n for circulant codes. Closing the four entries requires either codes meeting the Singleton bound, which would be AME stabilizer states, or improved upper bounds from linear or semidefinite programming. Even-length entries such as $[\![20, 0, d]\!]_5$ $(9 \le d \le 11)$, which lies below the $n = 18$ entry, also merit attention.

TABLE V. The four entries of Ref. [6] on two dates, from saved copies of the pages. "Modified" is the date printed on the page. A_d is the number of stabilizer elements of minimum weight d (d is the new lower bound), computed with the rank criterion of Sec. IV D for the code stored in the table and for ours.

	Page of 11 October 2026		Page of 9 November 2026		A_d	
Entry	Bounds, record	Modified	Bounds, record	Modified	Table	Ours
$[\![19, 0, d]\!]_5$	8–10, stored matrix	2014-10-16	9–10, cyclic, type "Gra"	2026-10-21	11 780	12 920
$[\![25, 0, d]\!]_5$	10–13, stored matrix	2014-10-16	11–13, cyclic, type "Gra"	2026-10-21	20 900	18 300
$[\![29, 0, d]\!]_5$	11–15, from stored $[\![30, 0, 12]\!]_5$ [a]	2014-10-16	12–15, cyclic, type "Gra"	2026-10-21	13 456	11 948
$[\![21, 0, d]\!]_7$	9–11, stored matrix	2014-10-23	10–11, cyclic, type "Gra"	2026-10-22	39 312	48 006

[a] Shortening to $[\![29, 1, 11]\!]_5$, then a subcode.

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DATA AVAILABILITY

The generators of all four codes are given in Tables I and II, which suffices to reproduce every distance and count stated here. The project repository (directory `problems/p01-code-tables`) contains the candidate files, the verifier with its test suite, the three independent programs of Sec. IV D, the rate-measurement program, the saved copies of the table pages of 10–11 October and 9 November 2026, the outputs of the runs quoted in Tables III, IV, and V, and a script (`results/replay.sh`) that repeats them. The times of run 1 in Table IV are taken from the submission database of the search. The validation on 14 stored table codes (Sec. IV) was not repeated for this revision.

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